MIL-C-43251C
8 February 1985
SUPERSEDING
MIL-C-43251B
23 February 1977

MILITARY SPECIFICATION

# CLOTH, PILE, SYNTHETIC FIBER PILE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This document covers knitted pile cloth.

1.2 Classification. The pile cloth shall be of the following classes as specified (see 6.2).

Class 1 - Green 252
Class 2 - Black 410
Class 3 - Green 394
Class 4 - Brown 463

## 2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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SPECIFICATION

FEDERAL

PPP-P-1133 - Packaging of Synthetic Fiber Fabrics

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MTL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

Laws and Regulations

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.



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## 3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample for shade, appearance, and hand and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 Pile.

3.2.1.1 Classes 1, 2, and 3. The pile of the cloth shall be acrylic fiber staple resulting in a pile matching that of the standard sample.

3.2.1.2 Class 4. The pile of the cloth shall be polyester fiber staple resulting in a pile matching that of the standard sample.

3.2.2 Backing.

3.2.2.1 Class 1. The backing for class 1 shall be made from undyed spun acrylic or polyester yarn.

3.2.2.2 Class 2. The backing for class 2 shall be made from dyed spun acrylic or polyester yarn, or fibrillated filament polypropylene yarn. The color of the backing yarn shall be oxford grey (see 6.7) or black. When fibrillated polypropylene yarn is used it shall be "doped" dyed black.

3.2.2.3 Class 3. The backing shall be made from undyed spun acrylic, polyester, fibrillated polypropylene, or cotton yarn.

3.2.2.4 Class 4. The backing shall be knitted from dyed polyester yarn. The color of the backing yarn shall be brown.

3.3 Color. Unless otherwise specified, the color of the pile shall be as listed below, and shall match the standard sample (see 6.3). The dullness (lack of lusture) shall approximate that of the standard sample.

Class 1 - Green 252
Class 2 - Black 410
Class 3 - Green 394
Class 4 - Brown 463

3.3.1 Matching. The color of the finished cloth shall match the standard sample when viewed from a position normal to the cloth surface and under artificial daylight having a color temperature of 7000 ± 500 K and the color shall be a good approximation to the standard sample under incandescent lamplight at 2850 ± 100 K. Because of directional effects, the standard and the sample to be compared shall be brushed and viewed in the same direction relative to construction and to angle of lighting.



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3.3.2 Colorfastness.

3.3.2.1 Colorfastness (class 1). The finished class 1 cloth shall show colorfastness to laundering equal to or better than the standard sample. When no standard sample has been established or designated as applicable to colorfastness, the finished cloth shall show good fastness to laundering when tested as specified in 4.2.3.

3.3.2.2 Colorfastness (classes 2 and 3). The finished class 2 and class 3 cloth shall show colorfastness to perspiration, laundering, dry cleaning and crocking equal to or better than the standard sample. When no standard sample has been established or designed as applicable to colorfastness, the finished cloth shall show good fastness to perspiration, laundering, dry cleaning and shall have a AATCC Chromatic Transference Scale rating for crocking not lower than 3.5 when tested as specified in 4.2.3.

3.3.2.3 Colorfastness (class 4). The finished class 4 cloth shall show colorfastness to perspiration, laundering, light and crocking equal to or better than the standard sample. When no standard sample has been identified, the finished cloth shall show good fastness to perspiration, laundering and light and shall have an AATCC Chromatic Transference Scale rating for crocking not lower than 3.5, when tested as specified in 4.2.3.

3.4 Physical requirements. The finished cloth shall conform to the requirements in table I when tested as specified in 4.2.3.

TABLE I. Physical requirements

| Characteristic | Requirement | | | | | | |
|---|---|---|---|---|---|---|---|
| | Class 1 | | Class 2 | | Class 3 | | Class 4 |
| | (Min) | (Max) | (Min) | (Max) | (Min) | (Max) | (Min) |
| Weight, ounces per square yard | 11.5 | 13.5 | 13.0 | - | 8.5 | 10.5 | 11.0 |
| Backing texture: | | | | | | | |
| Wales per inch | 10 | - | 10 | - | 20 | - | 15 |
| Courses per inch | 20 | | 20 | | 17 | | 22 |
| Shearable pile weight, ounces per square yard | 4.2 | - | 4.0 | - | 1.8 | - | 4.0 |
| Pile height, inches | 12/32 | 14/32 | 8/32 | 10/32 | 7/32 | 9/32 | 14/32 |
| Bursting strength, lbs | - | - | 60 | - | - | - | 190 |



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3.5 Finish. The cloth shall be opened and sheared. Pile characteristics shall be equal to those of the standard sample. An acrylic type resin applied to the back of the cloth as a bonding or anticurl agent will be permitted. The class 2 cloth shall be opened, sheared, and polished to the characteristic of a commercial "seal" finish.

3.6 Width. The minimum relaxed width of the finished cloth shall be as specified (see 6.2),

3.7 Dimensional stability (classes 2 and 4). The finished class 2 cloth shall not shrink more than 5.0 percent in length (wale direction) or width (course direction) and shall not elongate more than 1.0 percent in width after one cycle of dry cleaning, when tested as specified in 4.2.3. The finished class 4 cloth shall not shrink more than 5.0 percent in length (wale direction) or width (course direction) and shall not elongate more than 1.0 percent in width after laundering, when tested as specified in 4.2.3.

3.8 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 10 yards in the relaxed condition (see 4.2.2.3). Each length shall be put-up in a full width roll as specified in 5.1.

3.9 Fiber identification. Each roll shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.10 Workmanship. The finished cloth shall conform to the quality established by this document. The occurrence of defects shall not exceed the applicable quality levels specified herein.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.



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4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 Examination of the end item.

4.2.2.1 Yard-by-yard examination. The cloth shall be examined on the face side for the individual defects listed below. All defects found shall be counted regardless of their proximity to each other. The lot size shall be expressed in yards. The sample unit shall be 1 linear yard. The inspection level shall be I and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. The number of rolls, from which the sample yardage is to be selected shall be in accordance with table II. The sample yardage shall be apportioned equally among the selected rolls.

Individual defects

Cut, hole, or tear
Pile matted or glazed
Local area with pile of abnormal height or density
Creases or curled edges not removable by hand pressure
Spot, stain, or streak

4.2.2.2 Overall examination. The cloth shall be examined for the overall defects listed below. Each defect listed shall be counted not more than once in each roll examined. The sample size shall be the applicable number of rolls indicated in table II. Each roll in the sample shall be examined over its entire length. The lot shall be rejected if the total number of defects in the sample exceeds the acceptance number specified in table II.

Overall defects

Pile character or shade not matching that of the standard sample
Backing dyed for classes 1 and 3 or not specified color for class 2 and 4
Appreciable shedding of loose fiber
Dullness or color contrast not approximating that of the standard sample or mottled color



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TABLE II. Sample size

| Lot size (yards) | Sample size (rolls) | Acceptance number |
|---|---|---|
| 1,200 or less | 5 | 0 |
| 1,201 up to and including 3,200 | 8 | 0 |
| 3,201 up to and including 10,000 | 13 | 1 |
| 10,001 and over | 20 | 2 |

4.2.2.3 Length examination. Each roll of cloth used in the yard-by-yard examination shall be examined for length. Any length found to be less than the minimum specified or more than 1 yard below the length marked on the ticket shall be considered a defect with respect to length. All length determinations shall be based in the relaxed condition of the cloth. Measuring devices and conditions used shall be checked, and corrected as indicated, by lengths determined on cloth relaxed for 1 hour while loosely plaited on a smooth table top or similar surface, and then measured without re-rolling or tensioning. In case of dispute, measurement so made shall govern. If the total number of defects in the sample rolls exceeds the acceptance number specified in table II or if the total of the actual lengths of the sample rolls is less than the total of the lengths marked on the roll tickets, the lot shall be rejected.

4.2.2.4 Roll identification examination. During the yard-by-yard examination, each roll in the sample shall be examined for identification of fiber content. The lot shall be unacceptable if two or more rolls in the sample are not labeled or ticketed in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

4.2.3 End item testing. The cloth shall be tested for the applicable characteristics listed in table III. The methods of testing specified in FED-STD-191, wherever applicable and as listed in table III shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on the sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values used in expressing the final results. The sample unit shall be 1-1/2 yards full width of the cloth. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size shall be in accordance with the following:

| Lot size (yards) | Sample size (sample units) |
|---|---|
| 800 or less | 2 |
| 801 up to and including 22,000 | 3 |
| 22,001 and over | 5 |



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TABLE III. End item tests

| Characteristic | Requirement paragraph | Test method |
|---|---|---|
| Fiber identification | 3.2 | 1/ |
| Colorfastness: | | |
| Laundering (all classes) | 3.3.2 | 5614 |
| Perspiration (classes 2, 3, and 4) | 3.3.2 | 5680 |
| Dry cleaning (classes 2 and 3) | 3.3.2 | 5622 |
| Crocking (classes 2, 3, and 4) | 3.3.2 | 5651 |
| Light (class 4 only) | 3.3.2 | 5660 |
| Weight | 3.4 | 5041 |
| Backing texture | 3.4 | 5070 |
| Shearable pile weight | 3.4 | 5041 and 4.3.1 |
| Pile height | 3.4 | 4.3.2 |
| Bursting strength (classes 2 and 4) | 3.4 | 5120 |
| Dimensional stability: | | |
| Class 2 | 3.7 | 5580 |
| Class 4 | 3.7 | 5556 2/ |

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ Wool laundering procedure.

4.2.4 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1133.

4.3 Methods of inspection.

4.3.1 Test for shearable pile weight. The original (total) weight of the specimen shall be determined by Method 5041 using a specimen size not less than 4 by 4 inches square. The weighed specimen shall then be placed on a flat surface with the pile side up, and the pile shall be sheared off with an electric clipper (see 6.6). Shearing shall be done in the direction of the wales and in the direction of the courses, with the base of the cutting head parallel to the plane of the specimen. Following the shearing, the specimen shall be shaken to remove the shearings and shall be weighed. This procedure shall be repeated until the sheared backing attains a constant weight with 0.05 gram



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tolerance. The sheared backing shall be examined for damage to the backing yarns. If damage is found, the results shall be discarded and the entire procedure repeated. The weight of shearable pile shall be determined by subtracting the weight of the sheared backing from the original weight. Reporting shall be in accordance with Method 5041.

4.3.2 Pile height determination. The pile height shall be determined by use of a scale graduated to 1/32 inch or finer. The end of the scale shall be inserted vertically into the pile so that it rests squarely but without pressure on the backing yarns. The pile adjacent to the face of the scale shall be straightened, laid along the scale surface, and measured at the fiber ends. Five determinations shall be made at locations across and along the sample. The result shall be reported as the average of the five determinations expressed to the nearest 1/32 inch.

## 5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Levels A and Commercial. The batting shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Levels A, B and Commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1133.

## 6. NOTES

6.1 Intended use. The class 1 cloth is intended for use as the lining component in canteen covers. The class 2 cloth is intended for use in removable raincoat liners. The class 3 cloth is intended for use in the overcoat/raincoat removable liner. The class 4 cloth is intended for use in the fiberpile shirt of the Extended Cold Weather Clothing System (ECWCS).

6.2 Ordering data. Acquisition documents should specify the following:



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a. Title, number, and date of this document.
b. Class of cloth required (see 1.2).
c. Width of cloth required (see 3.6).
d. Minimum length if other than specified (see 3.8).
e. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample, address the contracting activity issuing the invitation for bids.

6.4 Dye formula (class 1). The acrylic pile for class 1 has been stock-dyed using cationic (basic) dyes. The following is a suggested but not mandatory dye formulation for Green 252:

Basic Blue 3G C.I. No. 51005
Basic Orange C.I. No. 48035
No. 21
Basic Red
No. 14

6.5 Class 1 cloth. The class 1 cloth is based on commercial cloth made from 10/1 (cotton count) backing yarn.

6.6 Electric clipper. An Oster Model (or equal) No. 111 carrying a No. 50, size 000 cutting head is recommended, Local barber supply agencies may be sources for the clipper described. A source is Oster Mfg, Co., 5055 N, Lydell Street, Milwaukee, WI.

6.7 Oxford grey (see 3.2.2.2). The oxford grey backing for class 2 has been successfully made using a minimum of 35 percent dyed black fiber and the remainder natural fiber.

6.8 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-C-43251C | Cloth, Pile, Synthetic Fiber Pile |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* — Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* — Optional |
|---|---|
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